

MSMS 401 : Bayesian Statistics and Computation

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1 Hierarchical Bayesian Models

A hierarchical Bayesian model (also called a **multilevel Bayesian model**) is a Bayesian model where parameters themselves are treated as random variables with their own prior distributions, and those priors may themselves depend on higher-level parameters.

A Hierarchical Bayesian model can be expressed in the form

$$p(\underline{x} | \boldsymbol{\theta}) = p(x_1, x_2, \dots, x_n | \boldsymbol{\theta}) = \prod_{i=1}^n p(x_i | \theta_i)$$

$$p(\boldsymbol{\theta} | \phi) = p(\theta_1, \theta_2, \dots, \theta_n | \phi) = \prod_{i=1}^n p(\theta_i | \phi)$$

$$\phi \sim p(\phi)$$

Observations $x_1, x_2, \dots, x_n$ are available from n different but related sources.

- The first stage of hierarchy specifies the parametric model component of the n observations.
- Since n observations are related, $\theta_1, \theta_2, \dots, \theta_n$ are assumed to be exchangeable. Hence, second and third stage of hierarchy provide prior on $\boldsymbol{\theta}$ and ϕ .

In many applications, it may be the interest to make inferences about unit characteristics — θ_i 's and population characteristics — ϕ . In either case, straight forward probability manipulations involving Bayes theorem provide the required posterior inferences.

$$p(\theta_i | \mathbf{x}) = \int p(\theta_i | \phi, \mathbf{x}) \cdot p(\phi | \mathbf{x}) d\phi$$

$$p(\theta_i | \phi, \mathbf{x}) \propto p(\mathbf{x} | \theta_i) \cdot p(\theta_i | \phi)$$

$$p(\phi | \mathbf{x}) \propto p(\mathbf{x} | \phi) \cdot p(\phi)$$

$$p(\mathbf{x} | \phi) \propto \int p(\mathbf{x} | \boldsymbol{\theta}) \cdot p(\boldsymbol{\theta} | \phi) d\boldsymbol{\theta}$$

Bayesian hierarchical modelling makes use of two important concepts while deriving posterior distribution —

- (i) Hyperparameters *i.e.* parameters of the prior distribution,
- (ii) Hyper-prior *i.e.* distribution of hyperparameters.

Example 1. Suppose $X | \mu \sim N(\mu, 1)$, $\mu | \theta \sim N(\theta, 1)$, $\theta \sim N(0, 1)$. Then the joint posterior of μ and θ can be given by

$$\begin{aligned} p(\mu, \theta | x) &\propto p(x | \mu) \cdot p(\mu | \theta) \cdot p(\theta) \\ &= \exp \left\{ -\frac{(x - \mu)^2}{2} - \frac{(\mu - \theta)^2}{2} - \frac{\theta^2}{2} \right\}. \end{aligned}$$

Now we can easily find the posterior inferences about μ and θ using the above results.

Example 2. Suppose $X | \lambda \sim \mathcal{P}(\lambda)$, $\lambda | \beta \sim \mathcal{G}(2, \beta)$, $\pi(\beta) \propto \frac{1}{\beta}$. Then,

$$\begin{aligned} p(x | \beta) &= \int_{\mathbf{R}^+} p(x | \lambda) \cdot p(\lambda | \beta) d\lambda \\ &= \int_{\mathbf{R}^+} \frac{e^{-\lambda} \lambda^x}{x!} \cdot \frac{\beta^2}{\Gamma(2)} e^{-\beta \lambda} \lambda^{2-1} d\lambda \\ &= \frac{\beta^2}{x!} \cdot \int_0^\infty e^{-(\beta+1)\lambda} \lambda^{x+2-1} d\lambda \\ &= \frac{\beta^2}{x!} \cdot \frac{\Gamma(x+2)}{(\beta+1)^{x+2}} \\ &= \frac{\beta^2}{x!} \cdot \frac{(x+1)!}{(\beta+1)^{x+2}} \\ &= \frac{\beta^2 \cdot (x+1)}{(\beta+1)^{x+2}} \end{aligned}$$

So, the conditional posterior of λ is

$$\begin{aligned} p(\lambda | \beta, x) &= \frac{p(x | \lambda) \cdot p(\lambda | \beta)}{p(x | \beta)} \\ &= \frac{\frac{e^{-\lambda} \lambda^x}{x!} \cdot \frac{\beta^2}{\Gamma(2)} e^{-\beta \lambda} \lambda^{2-1}}{\frac{\beta^2 \cdot (x+1)}{(\beta+1)^{x+2}}} \end{aligned}$$

$$\begin{aligned}
&= \frac{e^{-(\beta+1)\lambda} \cdot \lambda^{x+1} \cdot (\beta+1)^{x+2}}{(x+1)!} \\
&= \frac{(\beta+1)^{x+2}}{\Gamma(x+2)} \cdot e^{-(\beta+1)\lambda} \cdot \lambda^{x+1} \text{ i.e. } \mathcal{G}(2+x, \beta+1).
\end{aligned}$$

Recall that, Gamma prior is conjugate for Poisson parameter λ . So rightfully here

$$\lambda \mid \beta, x \sim \mathcal{G}(2+x, \beta+1).$$

Again, posterior of β upto proportionality is

$$\begin{aligned}
p(\beta \mid x) &\propto p(x \mid \beta) \cdot p(\beta) \\
&= \frac{\beta^2 \cdot (x+1)}{(\beta+1)^{x+2}} \cdot \frac{1}{\beta} \\
&= \frac{\beta \cdot (x+1)}{(\beta+1)^{x+2}}
\end{aligned}$$

Introducing the normalizing constant, the posterior of β is

$$\begin{aligned}
p(\beta \mid x) &= \frac{\beta \cdot (x+1)}{(\beta+1)^{x+2}} \bigg/ \int_0^\infty \frac{\beta \cdot (x+1)}{(\beta+1)^{x+2}} d\beta \\
&= \frac{\beta \cdot (x+1)}{(\beta+1)^{x+2}} \bigg/ (x+1) \cdot \mathcal{B}(2, x); \text{ Beta integration of second kind} \\
&= \frac{\beta \cdot (x+1)}{(\beta+1)^{x+2}} \bigg/ (x+1) \cdot \frac{\Gamma(2) \cdot \Gamma(x)}{\Gamma(2+x)} \\
&= \frac{\beta \cdot (x+1)}{(\beta+1)^{x+2}} \bigg/ (x+1) \cdot \frac{(x-1)!}{(x+1)!} \\
&= \frac{\beta \cdot (x+1)}{(\beta+1)^{x+2}} \bigg/ \frac{1}{x} \\
&= \frac{\beta \cdot (x+1) \cdot x}{(\beta+1)^{x+2}} \\
&= \frac{1}{\mathcal{B}(2, x)} \cdot \frac{\beta}{(\beta+1)^{x+2}} \text{ i.e. } \mathcal{B}(2, x).
\end{aligned}$$

$$\boxed{\therefore \text{ a posteriori } \beta \mid x \sim \mathcal{B}(2, x) .}$$

Finally, the marginal posterior of λ is

$$\begin{aligned}
 p(\lambda \mid x) &= \int_0^{\infty} p(\lambda, \beta \mid x) \, d\beta \\
 &= \int_0^{\infty} p(\lambda \mid \beta, x) \cdot p(\beta \mid x) \, d\beta \\
 &= \int_0^{\infty} \frac{e^{-(\beta+1)\lambda} \cdot \lambda^{x+1} \cdot (\beta+1)^{x+2}}{(x+1)!} \cdot \frac{\beta \cdot (x+1) \cdot x}{(\beta+1)^{x+2}} \, d\beta \\
 &= \frac{e^{-\lambda} \lambda^{x+1}}{(x-1)!} \int_0^{\infty} e^{-\lambda\beta} \beta \, d\beta \\
 &= \frac{e^{-\lambda} \lambda^{x+1}}{(x-1)!} \cdot \frac{\Gamma(2)}{\lambda^2} \\
 &= \frac{e^{-\lambda} \lambda^{x-1}}{\Gamma(x)}
 \end{aligned}$$

$$\boxed{\therefore \text{ a posteriori } \lambda \mid x \sim \mathcal{G}(x, 1).}$$